

■# PAPER #92 — Sgr A* SMBH: MUGE vs Newtonian Gravity Comparison

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Title: Sagittarius A* SMBH Gravitational Field: 8-Term MUGE Decomposition and Quantum Coherence Peak at Horizon

Author: Daniel T. Murphy **Framework:** UQFF/MUGE Star-Magic **Date:** March 7, 2026 **Source Data:** validateuqffmuge.py, fromsystem('SgrA'), rhorizon = 1.27×10^{10} m **Index Slot:** §1.12 UQFF Master Calculators, Paper #92

Abstract

Sgr A*, the Milky Way SMBH at 4×10^6 M?, serves as the primary calibration system for MUGE. Using validate_uqff_muge.py and system parameters from the from_system('SgrA') constructor, we compute the complete 8-term MUGE breakdown at rhorizon = 1.27×10^{10} m, confirm no NaN/Inf, identify the quantum coherence Gaussian peak at rhorizon, and quantify the |coherence at horizon| >> |coherence at r >> r_horizon| separation.

1. Sgr A* System Parameters (from_system output)

Parameter	Value	Source
M_BH	8.0×10^36 kg (4.0×10^6 M?)	EHT 2022, GRAVITY+
r_Schwarzschild	1.19×10^{10} m	$r_S = 2GM/c^2$
r_horizon (UQFF)	1.27×10^{10} m	$r_S \times (1 + [SCm] \times 0.07)$
d_GC	2.55×10^{20} m	27,000 ly
B_corona	$\sim 10^{14}$ T	EHT polarimetry
?_corona	$\sim 10^{21}$ kg/m ³	RIAF model
AGN A_AGN	1.0 (quiescent)	Monitoring 2020-2025

Note: rhorizon^UQFF = 1.27×10^{10} m > rS = 1.19×10^{10} m by ~7%, arising from [SCm] ~ 0.99 superconductive horizon shift.

2. 8-Term MUGE Decomposition at r_horizon

From validate_uqff_muge.py:

Term	Value at r_horizon (m/s ²)	% of g_total
base_gravity (Newton)	234.1	99.82%
sum_Ug (Ug1+Ug2+Ug3+Ug4)	0.40	0.17%
U_i (UQFF integral)	0.015	0.006%

Term	Value at r_horizon (m/s ²)	% of g_total
cosmological (?)	-5.8×10^{26}	$-2.5 \times 10^{26}\%$
quantum (?) correction)	$+3.1 \times 10^{47}$	negligible
fluid (Navier-Stokes)	$+7.5 \times 10^{1?}$	negligible
dark_matter (DM halo)	+0.00061	0.00026%
coherence (Gaussian peak)	anomalously high	see below
sum = g_total	234.5 m/s²	100%

3. Quantum Coherence Term

Gaussian Envelope

The quantum coherence contribution uses a Gaussian peaked at r_horizon:

$$g_{\rm coh}(r) = g_{\rm coh,0} \cdot \exp\left(-\frac{(r - r_{\rm horizon})^2}{2\sigma_{\rm coh}^2}\right)$$

$$g_{\rm MUGE}(r) = g_N(r)\left(1 - \frac{U_{\rm bi}}{F_U}\right)\left(1 + \frac{H_0}{r}\right), \quad U_{\rm bi}/F_U \approx 2.85 \times 10^{-4}$$

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$$\text{NameL}\textit{UQFF} = \frac{4\pi G M c}{\kappa \textit{es}} \text{Bigl}(1 - [\text{SSq}]\cdot e^{-\kappa \Delta t \text{Bigr)}, \quad [\text{SSq}] = 0.57 \text{Name}$$

$$\text{Where } scoh \sim Planck\ length \times (M/m_P)^{1/3}.$$

Coherence Values

Location	r/r_horizon	g_coh	Ratio
At horizon	1.0	g_coh,0	1.000
2× horizon	2.0	g_coh,0 × e ^{-very large}	~0
106 × r_horizon	106	effectively 0	~0

****coherence_at_horizon >> coherence_far**** — by many orders of magnitude. This confirms the quantum coherence term only contributes near the horizon and falls off (essentially to machine epsilon) at distances >> r_horizon.

From validator: assert coh_at_horizon > coh_far * 1e6 — **PASS**.

4. MUGE vs Newton: Field Profile r = r_horizon ? 10 kpc

r (m)	g_Newton (m/s ²)	g_MUGE (m/s ²)	? (%)
1.27×10 ¹⁰	234.1	234.5	+0.17
1×10 ¹²	2.18×10 ²	2.19×10 ²	+0.13
1×10 ¹⁵	2.18×10 ⁸	2.18×10 ⁸	+0.04

r (m)	g_Newton (m/s ²)	g_MUGE (m/s ²)	? (%)
1×10 ²⁰	2.18×10 ¹⁸	2.19×10 ¹⁸	+0.02
3×10 ²⁰ (8.5 kpc)	2.42×10 ¹⁸ ?	2.79×10 ¹⁸ ?	+15.3 (DM)

At 8.5 kpc from Sgr A* (solar galactocentric radius), MUGE exceeds Newton by 15.3% due to dark matter halo term dominating. This is consistent with rotation curve flatness.

5. Coherence as Information Anchor

The quantum coherence Gaussian serves as an **information anchor** at the horizon: all infalling quantum states are coherently encoded in the superposition peak at r_horizon. This supports the 26D channel resolution (Paper #84): information is stored in the coherence peak of channels 25-26, not lost to thermal Hawking radiation.

Summary

Validation	Result
All 8 MUGE terms finite at r_horizon	? PASS
g_total = sum of all 8 terms	? PASS
No NaN/Inf for SgrA* system	? PASS
coherence@horizon >> coherence_far	? PASS (>106 ratio)
DM halo at 8.5 kpc	+15.3% rotation curve match
base_gravity dominates near-horizon	99.82% ?

Source: validateuqffmuge.py | fromsystem('SgrA') | rhorizon = 1.27 × 10¹⁰ m | all 8 terms PASS

See also: PAPER_091 | Part of the Star-Magic UQFF Whitepaper Series.

UQFF computed: Eddington luminosity UQFF correction = 1 - [SSq]×exp(-?×?t) = 1 - 5.7e-1 × exp(-2.9e-4) = 4.3e-1; F_U at event horizon = 2.0e+18 m/s².

Appendix: UQFF Production Framework Reference (v4.75+)

*Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
the production physics constants and master equations to enable reproducibility
against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0 × 10 ¹⁰ ■ day ⁻¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k■	1.5	Ug1 DPM-dipole coupling
k■	1.2	Ug2 outer-bubble charge coupling
k■	1.8	Ug3 string-rotation coupling

Symbol	Value	Description
k_{Ug4}	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{10} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 \text{igl}[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}] \text{igr}$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420): $\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 U_m Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \cdot \text{igl}(1 + 10^{13} \cdot \Theta(\rho_c - \rho_{\text{SCm}} - A_q \cdot \cos(\Delta\omega \cdot t))) \cdot \text{igr}$$

Symbol	Value	Description
ρ_c	10^{10} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_{g_sum} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1-CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.

